

Offline Lead Management – Technical KT & Feasibility Notes

1. Existing Architecture Overview

The current solution consists of three major components:

1. HubSpot → Main Google Sheet

- HubSpot workflows push data into the **Main Dashboard (Leads Data)**.
 - Multiple HubSpot workflows update different fields independently.
 - The Main Dashboard acts as the source of truth for all reseller dashboards.
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2. Main Dashboard → Reseller Dashboards

- Implemented using Google Apps Script.
 - A scheduled trigger executes every **1 minute**.
 - The script:
 - Reads rows from the Main Dashboard.
 - Determines the reseller using the Reseller Name column.
 - Creates reseller dashboards automatically if they do not exist.
 - Inserts or updates records in the corresponding reseller dashboard.
 - Tracks synchronization status using **Column T**.
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3. Google Sheets → HubSpot

- Implemented using **Superjoin**.
 - Runs automatically every **6 hours**.
 - Updates only selected HubSpot properties.
 - No custom Apps Script is involved in this flow.
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2. Superjoin KT

Current Understanding

Superjoin is responsible only for:

- Exporting data from Google Sheets to HubSpot.
- Scheduled synchronization every 6 hours.
- Field mapping.

- Automation configuration.

No custom code exists for Superjoin inside Apps Script.

Developer Scope

Current developer ownership is limited to:

- Google Apps Script
- HubSpot integrations where scripting is required

Superjoin automation configuration itself is not part of the Apps Script implementation.

Investigation

During KT:

- Queries were located successfully.
- Active exports were verified.
- Automation was confirmed to be running.

However:

- Export schedule could not be modified.
- Query editor remained unavailable.

This appears to be a Superjoin configuration or permission issue rather than an Apps Script issue.

3. Existing Apps Script Analysis

Current synchronization logic:

```
Main Dashboard
  ↓
Every minute
  ↓
Read all rows
  ↓
Find reseller
  ↓
Create dashboard if required
  ↓
Insert / Update row
```

↓
Mark Column T as synced

Column usage:

- A → Record ID
- S → Reseller Name
- T → Last Synced Reseller

4. Client Issues (A1-A3)

A1 – Updates not propagated

Current Behaviour

Once a row has been synchronized:

Column T = Reseller Name

Future executions skip the row:

```
if (t === s)
  continue;
```

Therefore later edits are never propagated.

Proposed Fix

Use Record ID based synchronization.

Workflow:

```
Row edited
  ↓
Capture Record ID
  ↓
Locate row
  ↓
```

Copy latest row



Overwrite reseller sheet

No dependency on Column T for updates.

Questions

- Which columns should trigger synchronization?
 - Should every field update trigger a sync?
 - Should HubSpot updates and manual edits both be synchronized?
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A2 – Deleted rows remain in reseller sheets

Current Behaviour

Deletion only happens when reseller assignment changes.

If a row disappears from the Main Dashboard:

- reseller sheet still retains it.
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Challenge

After deleting a row:

- Record ID no longer exists.
 - Apps Script cannot determine which record disappeared.
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Proposed Solution

Instead of physical deletion:

Introduce property:

Is Deleted

Yes / No

Workflow:

```
User marks Yes
↓
Apps Script detects change
↓
Fetch Record ID
↓
Delete reseller record
↓
Delete main record
```

Questions

- Should deletion happen immediately?
- Should a delay (24 hours) be maintained?
- Who will mark the deletion flag?

A3 – Rows marked synced before confirmation

Current Behaviour

Current code:

```
tVals[i][0] = s;
```

executes immediately after write.

No validation exists confirming successful write.

Proposed Fix

Only update Column T after successful completion of:

```
Insert / Update
```

If write fails:

```
Do not mark synced.
```

Questions

- What should happen when write fails?
 - Should retries be attempted?
 - Should failures be logged?
-

5. Estimated Effort

Current estimate:

10–15 Hours

Includes:

- Debugging
- Development
- Testing
- Validation

Estimated completion:

Approximately one week

6. Two-Way Synchronization Requirement

New requirement identified after discussion with Jyoti.

Current flow:

```
graph TD
    Main --> Reseller
```

New flow:

```
graph TD
    Main --> Reseller
    Reseller --> Main
```

Reseller changes should also update the Main Dashboard.

Feasibility

Possible approach:

```

Edit in reseller sheet
    ↓
Capture Record ID
    ↓
Find row in Main Dashboard
    ↓
Overwrite latest values

```

Additional design analysis is still required.

This requirement may change overall effort estimation.

7. Trigger Investigation

Several experiments were conducted.

On Edit

Successfully captured:

- Row
- Column
- Old Value
- New Value
- Record ID

Record ID is retrieved from Column A.

Example:

```

Record ID
↓

Find row

↓

```

Copy latest values

On Change

Tested:

changeType

Deleting rows consistently returned:

EDIT

instead of

REMOVE_ROW

Therefore Apps Script cannot reliably detect row deletions using change events.

8. Final Synchronization Approach

Updates

```
Edit
  ↓
Capture Record ID
  ↓
Locate Main Row
  ↓
Overwrite Reseller Row
```

Deletions

Recommended approach:

```
Is Deleted = Yes
    ↓
Capture Record ID
    ↓
Delete reseller record
    ↓
Delete main record
```

9. New Reseller Dashboard Apps Script

Future requirement:

Each reseller dashboard should contain its own Apps Script responsible for:

```
Reseller
    ↓
Main Dashboard
```

Current Main Dashboard already contains:

```
Main
    ↓
Reseller
```

Current Observation

New reseller dashboards are created automatically.

However they do not receive Apps Script automatically.

Possible solutions:

Preferred

Maintain a template reseller dashboard containing:

- Apps Script
- Required sheets
- Configuration

Create new dashboards by copying the template.

10. Region-Based Folder Structure

New feasibility request.

Initial implementation:

Region:

GDE

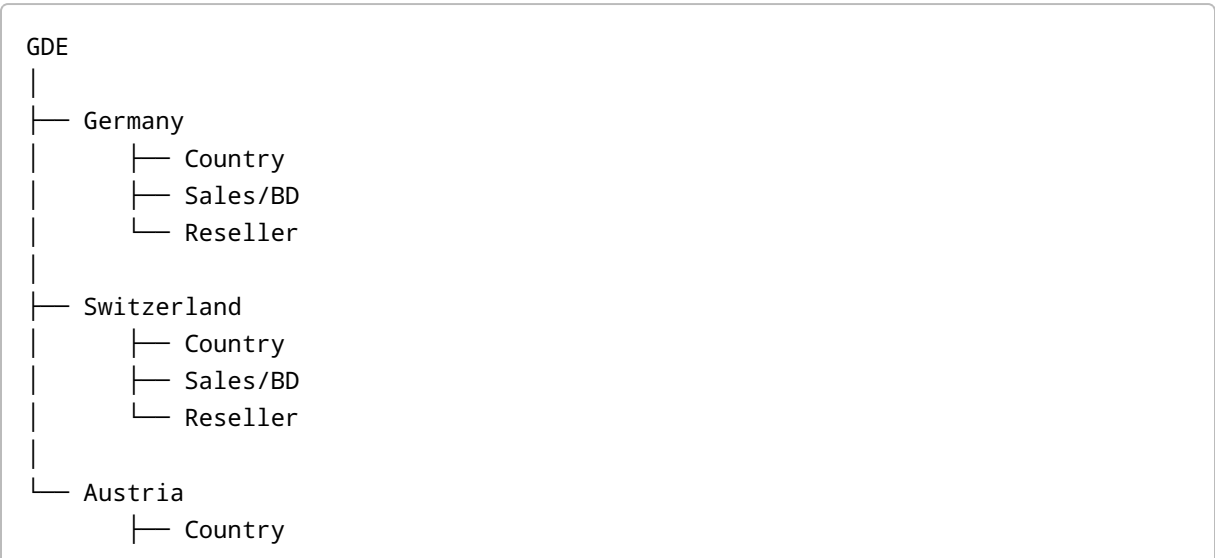
Countries:

- Germany
- Switzerland
- Austria

Each country contains:

Country
Sales/BD
Reseller

Folder Structure



└ Sales/BD
└ Reseller

Proposed Logic

Main Dashboard already contains:

- Country
- Reseller

App Script will maintain a hardcoded mapping:

Germany

↓

GDE

Folder IDs will also be maintained inside the script.

Workflow:

Read Country

↓

Determine Region

↓

Locate Folder

↓

Create required dashboards

Feasibility Questions

Before implementation:

- Are folders already created?
- Should Apps Script create folders automatically?
- Do all three dashboard types have identical structures?
- Does Country contain only offline leads?
- Does Reseller contain online + offline leads?
- Should existing dashboards be migrated?
- Should edits from all dashboards sync back to Main?

11. Overall Technical Direction

Current architecture remains viable.

Primary enhancements under consideration:

- Reliable update synchronization using Record ID.
- Safe deletion workflow using a soft delete flag.
- Two-way synchronization between reseller dashboards and Main Dashboard.
- Template-based reseller dashboard creation.
- Region and country based dashboard organization.
- Future scalability without redesigning the current synchronization engine.

These changes extend the existing architecture rather than replacing it and are considered technically feasible with moderate development effort.